Given values for these two parameters, R-trees abide by a few invariants:

1. Every leaf contains between `m` and `M` points (except for the root, which can possibly have less than `m` points);
2. Each leaf node L has associated a hyper-rectangle `RL`, such that `RL` is the smallest rectangle containing all the points in the leaf.
3. Every internal node has between `m` and `M` children (except for the root, which can possibly have less than `m` children);
4. Each internal node `N` has associated a bounding (hyper-)rectangle `RN`, such that `RN` is the smallest rectangle, whose edges are parallel to the cartesian axes, entirely containing all the bounding rectangles of `N`'s children;
5. The root node has at least 2 children, unless it is a leaf;
6. All leaves are at the same level.

Property number 6 tells us that R-trees are balanced, while from properties 1 and 3 we can infer that the maximum height of an R-tree containing `n` points is is `logm(n)`.

On insertion, if any node on the path from the root to the leaf holding the new point becomes larger than `M` entries, we will have to split it, creating two nodes each with half the elements.